

GRADUATION PROJECT PROPOSAL

Program	Computer Science / Artificial Intelligence
Project Title	Real-Time Weapon Detection in Grayscale CCTV Surveillance Using Deep Learning
Student Name(s) & ID(s)	Mohamed Tarek Abdou Mohamed Wael Mohamed Mohamed Kamal Abdel Fatah Mahmoud Ahmed Mohamed Youssef Mohamed Abdel Samie Al-Komi
Supervisor	Dr. Tarek Ghoniemy
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1. Abstract

Public safety infrastructure worldwide increasingly relies on closed-circuit television (CCTV) surveillance networks to detect and deter violent threats. However, the effectiveness of such systems is severely constrained by the cognitive limitations of human monitoring operators. Empirical research has established that human operators lose awareness of up to 95% of on-screen activity after only 20 to 40 minutes of sustained monitoring — a phenomenon known as 'video blindness' — resulting in critically delayed or altogether absent threat responses.

This proposal describes the design and development of a Real-Time Automated Weapon Threat Detection System leveraging the YOLOv8 (You Only Look Once, version 8) deep learning architecture. The system will be trained on a diverse weapon image dataset sourced from Roboflow Universe (weapon-detection-jqd3x) and will be engineered to accurately detect and localize firearms and bladed weapons in live surveillance footage with minimal inference latency. The project will encompass the complete machine learning pipeline: data acquisition and preprocessing, model training with augmentation strategies, performance evaluation against standard detection metrics, and deployment optimization for edge computing hardware. The expected output is a functional, high-speed detection prototype capable of generating automated alerts upon weapon identification, thereby reducing

reliance on human vigilance and significantly improving the temporal accuracy of threat response in public and institutional security environments.

2. Introduction and Background

The proliferation of weapon-related violence in public spaces — including schools, transit hubs, commercial centers, and government facilities — has created an urgent imperative for intelligent, automated surveillance infrastructure. While the global deployment of CCTV camera networks has expanded dramatically over the past two decades, the security value of these networks is fundamentally constrained by their dependence on human operators to interpret video feeds in real time.

Research in the field of human factors has repeatedly demonstrated that sustained visual vigilance is physiologically unsustainable. Stainer et al. (2013) documented that human monitoring performance degrades precipitously after 20 to 40 minutes of intensive screen observation, with operators failing to detect up to 95% of notable events. This limitation, compounded by the inherent difficulty of distinguishing weapons from visually similar innocuous objects in dynamic surveillance environments, renders traditional manual surveillance systematically insufficient for reliable threat detection.

The emergence of deep learning-based object detection, and in particular the YOLO (You Only Look Once) family of single-stage convolutional neural network architectures, has opened a viable pathway for automated weapon detection systems (WDS) capable of operating in real time. YOLOv8, the most recent and advanced iteration of this architecture as of this proposal, offers a state-of-the-art combination of detection accuracy, inference speed, and architectural flexibility, making it particularly well-suited for integration into live surveillance pipelines. This project situates itself within the broader context of smart city safety infrastructure and proposes to harness YOLOv8 for the development of a robust, deployable WDS prototype.

3. Problem Statement

Contemporary security surveillance systems face a dual-layered failure mode that collectively undermines their capacity to serve as reliable threat detection instruments. First, at the human level, operator fatigue and attentional decay — collectively described as 'video blindness' — mean that surveillance personnel cannot maintain the sustained vigilance required for consistent threat identification in large, multi-camera deployments. This is not a training or motivational deficiency; it is an irresolvable constraint of human cognitive architecture.

Second, at the algorithmic level, automated weapon detection presents a set of non-trivial computer vision challenges that have limited the effectiveness of prior generation detection systems:

- **Small Object Detection:** Firearms and bladed weapons frequently occupy only a small number of pixels within wide-angle surveillance frames, placing them at or below the effective resolution threshold of standard object detection feature extractors.
- **Occlusion and Background Clutter:** Weapons are routinely obscured by clothing, bags, or body parts, and surveillance environments are typically visually cluttered, elevating rates of both false positive and false negative detections.
- **Environmental Variability:** Real-world deployment involves fluctuating lighting conditions, motion blur, lens glare, and nighttime scenarios that significantly degrade model performance.
- **Dataset Scarcity:** The availability of diverse, high-quality, real-world weapon surveillance datasets remains limited, with many existing benchmark datasets composed of synthetic or cinematographic imagery that does not generalize reliably to authentic CCTV footage.

The convergence of these challenges means that no existing deployed system achieves the combination of accuracy, speed, and robustness required for unattended, real-time operation. This project directly addresses this gap through the application of YOLOv8, augmented with targeted preprocessing and augmentation strategies suited to the weapon detection domain.

4. Aim and Objectives

4.1 Aim

The overarching aim of this project is to design, develop, and evaluate a real-time automated weapon threat detection system based on the YOLOv8 architecture, capable of accurately identifying and localizing firearms and bladed weapons in live surveillance video feeds with sufficient speed and precision for deployment in operational security environments.

4.2 Objectives

1. To acquire and curate a representative weapon image dataset from the Roboflow Universe platform (weapon-detection-jqd3x) encompassing diverse weapon classes, environmental conditions, and viewpoints.
2. To preprocess and augment the acquired dataset using illumination normalization, geometric transformation, and synthetic data augmentation techniques to improve model generalization and address class imbalance.
3. To configure, fine-tune, and train a YOLOv8 model on the processed dataset, optimizing hyperparameters to balance detection accuracy and inference latency.
4. To evaluate model performance using standard object detection metrics including mean Average Precision (mAP), Precision, Recall, and Frames Per Second (FPS), benchmarking against published baselines.

5. To implement a false positive mitigation strategy incorporating temporal consistency filtering and hard negative mining to reduce erroneous weapon classifications in ambiguous scenes.
6. To develop a real-time video inference pipeline using OpenCV that integrates the trained YOLOv8 model with a live or pre-recorded surveillance feed and triggers automated visual alerts upon weapon detection.
7. To optimize the trained model for deployment on resource-constrained edge hardware through model compression techniques, including weight pruning and quantization, targeting a minimum throughput of 30 frames per second.
8. To document the system architecture, experimental results, and deployment procedure in a comprehensive technical report suitable for academic submission and operational handover.

5. Significance of the Project

This project carries significance across academic, practical, and social dimensions. Academically, it contributes an empirically evaluated YOLOv8 fine-tuning methodology tailored to the specific visual challenges of weapon detection in surveillance contexts, including findings on optimal augmentation strategies, hyperparameter configurations, and edge deployment trade-offs. These results will be relevant to the computer vision, physical security, and smart city research communities.

Practically, the developed system prototype addresses a demonstrable operational gap in existing security infrastructure by replacing or augmenting human operators as the primary threat identification mechanism. The system's alert generation capability can be directly integrated with existing security control rooms, law enforcement Computer-Aided Dispatch (CAD) systems, or building management platforms through standard API interfaces.

Socially, by enabling faster and more reliable automated detection of weapon threats, the system has the potential to reduce the response time of security and law enforcement personnel in active threat situations — a factor with direct implications for the preservation of human life. Beneficiaries of the developed system include airport and transit authorities, educational institutions, healthcare facilities, government buildings, and any public venue reliant on CCTV-based security infrastructure.

6. Scope and Limitations

6.1 Scope

- The project will focus on the detection of two primary weapon categories: firearms (handguns and long guns) and bladed weapons (knives), as represented in the selected Roboflow dataset.
- The detection system will operate on RGB visible-light imagery consistent with standard CCTV camera output; thermal or radar sensor inputs are outside the scope of this project.

- Model training and primary evaluation will be conducted on GPU-equipped workstations using the PyTorch framework.
- Edge deployment optimization will target NVIDIA Jetson-class hardware as the reference platform.
- The alert system will be implemented as a visual overlay and frame-level log; integration with external dispatch or law enforcement systems is outside the scope of the current proposal.

6.2 Limitations

- Detection performance may degrade in extreme low-light or heavy occlusion scenarios not adequately represented in the training dataset.
- The system does not incorporate behavioral analysis or intent assessment; it detects weapon presence only, not the threat context.
- The dataset sourced from Roboflow Universe may contain demographic or environmental biases that limit generalization to specific deployment geographies.
- Privacy and ethical considerations related to mass surveillance infrastructure are acknowledged but fall outside the technical scope of this proposal.

7. Literature Review / Related Work

7.1 Classical Machine Learning Approaches

Early automated weapon detection systems were built on hand-crafted feature representations. Dalal and Triggs (2005) introduced the Histogram of Oriented Gradients (HOG), which became a foundational descriptor for object detection tasks, typically paired with Support Vector Machine (SVM) classifiers. Lowe (2004) proposed the Scale-Invariant Feature Transform (SIFT), providing robustness to scale and rotation. While these methods achieved measurable success in controlled laboratory conditions, their brittleness under real-world variability — including illumination changes, partial occlusion, and viewpoint variation — rendered them insufficient for the demands of live surveillance systems.

7.2 Deep Learning and Convolutional Neural Networks

The introduction of deep Convolutional Neural Networks (CNNs) fundamentally transformed object detection. Two principal architectural families emerged: two-stage detectors and one-stage detectors. Two-stage architectures, most prominently Faster R-CNN (Ren et al., 2015), decompose detection into a region proposal stage followed by a classification and refinement stage. This approach achieves superior localization precision and supports instance-level segmentation (He et al., 2017), but at a significant computational cost that renders these architectures generally impractical for real-time video inference without dedicated hardware acceleration.

One-stage detectors process the entire image in a single forward pass, prioritizing inference speed. Redmon and Farhadi (2018) proposed YOLOv3, which established the YOLO family

as the dominant paradigm for real-time detection tasks. Subsequent versions have progressively addressed early YOLO limitations in small object detection through refined backbone architectures and multi-scale feature pyramid fusion. Salazar González et al. (2020) directly applied YOLO to CCTV weapon detection, demonstrating the viability of this architectural family for the surveillance domain.

7.3 Accuracy Enhancement Techniques

Recent research has applied targeted accuracy enhancement strategies to weapon-specific detection. Attention mechanisms, including the Convolutional Block Attention Module (CBAM) and Squeeze-and-Excitation blocks, enable networks to selectively amplify weapon-relevant features while suppressing background activations — a particularly valuable property in visually cluttered surveillance frames. Ensemble fusion strategies, such as Weighted Box Fusion (WBF) (Solovyev et al., 2021), aggregate predictions from multiple models to improve recall and reduce spurious detections.

7.4 Data Augmentation and Preprocessing

Given the persistent scarcity of large-scale, real-world weapon datasets (Salazar González et al., 2020), data augmentation plays a critical role in WDS development. Generative Adversarial Networks (Goodfellow et al., 2014) have been employed to synthesize photorealistic weapon images for training set expansion. Illumination normalization techniques have been applied to standardize frame appearance across varying lighting conditions, while hard negative mining explicitly exposes models to visually confusable non-weapon objects to reduce false positive rates.

7.5 Deployment and Emerging Directions

Li et al. (2021) highlighted the importance of edge computing for real-time surveillance applications, demonstrating that on-device inference on platforms such as the NVIDIA Jetson Nano can achieve latency requirements unachievable through cloud-reliant architectures. EfficientDet (Tan et al., 2020) and MobileNet variants offer scalable efficiency for deployment on resource-constrained hardware. Emerging research directions identified in the literature include Vision Transformer (ViT)-based architectures for improved global context reasoning (Dosovitskiy et al., 2020), multimodal sensor fusion incorporating thermal and radar data, and Federated Learning for privacy-preserving collaborative model training across distributed camera networks.

7.6 Research Gap

Despite the breadth of existing research, a persistent gap exists between laboratory detection benchmarks and robust, generalized real-world deployment. YOLOv8, as the most architecturally advanced member of the YOLO family, has not yet been comprehensively

evaluated specifically for the weapon detection surveillance domain with end-to-end pipeline optimization including edge deployment. This project directly addresses this gap.

8. Proposed Methodology

Phase 1: Data Acquisition

The primary training dataset will be sourced from the Roboflow Universe public repository, specifically the weapon-detection-jqd3x dataset (<https://universe.roboflow.com/ir-uz1qh/weapon-detection-jqd3x>). This dataset provides annotated images of firearms and bladed weapons across diverse environmental conditions. The dataset will be exported in YOLOv8-compatible format (YOLO annotation standard). A stratified train/validation/test split of 70%/15%/15% will be applied to ensure class distribution consistency across all splits.

Phase 2: Preprocessing and Augmentation

Raw images will undergo a standardized preprocessing pipeline prior to training. Illumination normalization will be applied to mitigate lighting variability across frames. The following augmentation techniques will be applied during training to improve model generalization and address class imbalance: random horizontal and vertical flipping; rotation and affine transformations; brightness, contrast, and saturation jitter; mosaic augmentation (a standard YOLOv8 training technique combining four images); and cutout/mixup regularization. Hard negative mining will be applied in later training epochs to explicitly reduce false positive rates on visually confusable objects.

Phase 3: Model Configuration and Training

YOLOv8 (Nano or Small variant, selected based on preliminary performance benchmarks) will be initialized with ImageNet pre-trained weights and fine-tuned on the weapon detection dataset. Training will be conducted using the Ultralytics YOLOv8 Python framework on a GPU-equipped workstation. A hyperparameter optimization sweep will be conducted over learning rate, batch size, and augmentation parameters to identify the configuration that maximizes validation mAP while maintaining target inference speed. Training will be monitored using loss curves, mAP@0.5, and mAP@0.5:0.95 metrics.

Phase 4: Evaluation

The trained model will be evaluated on the held-out test set using the following metrics: mean Average Precision at IoU 0.5 (mAP@0.5) and mAP@0.5:0.95; per-class Precision and Recall; F1 score; and Frames Per Second (FPS) on both GPU and edge hardware. Results will be compared against published baselines from the weapon detection literature to contextualize the system's performance.

Phase 5: Real-Time Inference Pipeline

The trained model will be integrated with an OpenCV-based video capture and display pipeline capable of processing live camera feeds or pre-recorded CCTV footage. The pipeline will apply temporal consistency filtering — requiring a positive detection to persist across a minimum of three consecutive frames before triggering an alert — to reduce transient false positives. Detected weapon instances will be annotated on the video frame with bounding boxes, class labels, and confidence scores. Alert events will be logged to a structured output file with timestamp and frame reference.

Phase 6: Edge Deployment Optimization

The trained PyTorch model will be exported to ONNX format and subsequently converted to TensorRT for deployment on NVIDIA Jetson hardware. Model compression will be applied through post-training quantization (INT8), targeting a minimum throughput of 30 FPS on the target edge platform. Final performance benchmarks will compare accuracy retention against the full-precision baseline to quantify the accuracy-efficiency trade-off.

9. Tools, Technologies, and Resources

Programming Language

- Python 3.10+

Deep Learning Framework

- PyTorch (model training and optimization)
- Ultralytics YOLOv8 (model architecture, training pipeline, and export utilities)

Computer Vision

- OpenCV (video capture, frame preprocessing, and real-time inference pipeline)

Data Management

- Roboflow Universe (weapon-detection-jqd3x dataset) — <https://universe.roboflow.com/ir-uz1qh/weapon-detection-jqd3x>
- Roboflow Python SDK (dataset versioning, export, and augmentation)

Model Optimization and Deployment

- ONNX Runtime (cross-platform model export)
- TensorRT (NVIDIA GPU and Jetson inference optimization)

Hardware

- GPU-equipped training workstation (NVIDIA GPU, minimum 8 GB VRAM)
- NVIDIA Jetson Nano or equivalent (edge deployment target)

Experiment Tracking

- Weights & Biases (WandB) or TensorBoard (training monitoring and metric logging)

Development Environment

- Jupyter Notebook / VS Code
- Git / GitHub (version control)

10. Expected Deliverables

- A fully annotated, preprocessed, and augmented training dataset derived from the Roboflow weapon-detection-jqd3x source, in YOLOv8-compatible format.
- A trained and evaluated YOLOv8 weapon detection model with documented performance benchmarks (mAP, Precision, Recall, FPS).
- A real-time video inference application integrating the trained model with an OpenCV pipeline, capable of processing live or pre-recorded surveillance footage and generating automated weapon detection alerts.
- An edge-optimized model (quantized, TensorRT-compiled) with deployment documentation for NVIDIA Jetson hardware.
- A comprehensive technical report documenting the methodology, experimental results, performance analysis, and conclusions.
- A project poster and final oral presentation summarizing the system design, results, and contributions.

11. Work Plan and Timeline

No.	Project Phase	Planned Duration
1	Literature Review — Survey of existing WDS, deep learning architectures, and grayscale preprocessing methods	Week 1 ✓ (Done)
2	Proposal Writing — Drafting and finalizing the graduation project proposal document	Week 2 ✓ (Done)
3	Dataset Acquisition — Download and validate weapon dataset from Roboflow Universe (weapon-detection-jqd3x); perform class distribution and quality analysis	Week 3
4	Data Preprocessing — Annotation verification, train/validation/test split, format conversion, and initial augmentation pipeline	Week 4
5	Grayscale Pipeline — Convert dataset to grayscale; design and validate the full grayscale preprocessing and inference pipeline	Week 5
6	GKDPHE Implementation — Implement and integrate Gaussian Kernelized Double Plateau Histogram Equalization for illumination normalization on grayscale frames	Week 6

7	YOLOv8 Setup — Configure YOLOv8 architecture, environment, and training infrastructure; verify GPU compatibility	Week 7
8	RGB Baseline Training — Train YOLOv8 on RGB dataset to establish performance baseline (mAP, Precision, Recall, FPS)	Week 8
9	Grayscale Training — Fine-tune YOLOv8 on GKDPHE-processed grayscale dataset; monitor convergence and validation metrics	Week 9
10	Hyperparameter Tuning — Systematic sweep over learning rate, batch size, augmentation, and anchor parameters for both RGB and grayscale models	Week 10
11	Evaluation — Comprehensive testing on held-out test set; compute mAP@0.5, mAP@0.5:0.95, Precision, Recall, F1, and FPS benchmarks	Week 11
12	Comparison — Quantitative and qualitative comparison between RGB baseline and grayscale GKDPHE model; false positive analysis	Week 12
13	Optimization — Apply model compression (quantization/pruning) for edge deployment; benchmark on target hardware	Week 13
14	Final Report & Presentation — Complete technical documentation, project poster, and oral presentation preparation	Week 14

12. Conclusion

This proposal has presented a well-grounded and technically feasible plan for the development of a Real-Time Automated Weapon Threat Detection System using the YOLOv8 deep learning architecture. The project is motivated by a clearly documented and persistent deficiency in contemporary surveillance security infrastructure: the fundamental inability of human operators to maintain sustained visual vigilance across large camera networks, and the absence of a widely deployed automated solution that effectively addresses the specific computer vision challenges of weapon detection — namely small object size, occlusion, environmental variability, and dataset scarcity.

By leveraging YOLOv8's state-of-the-art balance of detection accuracy and inference speed, combined with a rigorous data preprocessing and augmentation pipeline, a temporal consistency filtering mechanism, and an edge deployment optimization strategy, the proposed system is positioned to deliver a meaningful and measurable improvement over both manual monitoring and prior automated detection approaches. The project's deliverables — a trained and evaluated model, a functional real-time inference application,

and a comprehensive technical report — constitute a substantive academic and practical contribution to the field of intelligent surveillance and public safety systems.

The work is scoped realistically within the constraints of a graduation project timeline, with a phased plan that ensures progressive and verifiable milestones. The expected outcomes will serve as a solid foundation for future research into behavioral analytics, multimodal sensor fusion, and federated learning for distributed weapon detection networks.

13. References

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